



STANDARD OPERATING PROCEDURE	
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Title: Computer System Validation

1 PURPOSE

This Standard Operating Procedure (SOP) outlines the procedure required for initial validation and maintenance of the validated state in computerized systems used by Netramind for GxP activities.

2 SCOPE

This procedure applies to computerized systems used by Netramind in support of GxP-regulated activities, including clinical data analysis workflows and related supporting systems where validation is determined to be appropriate based on system risk and intended use.

3 REFERENCES

- [SOP-015 21 CFR Part 11](#)
- [SOP-016 Data Integrity](#)
- [SOP-004 GxP Records Retention](#)
- GAMP 5—A Risk-Based Approach to Compliant GxP Computerized Systems
- International Conference on Harmonization (ICH) E6 (R3) Good Clinical Practice
- International Conference on Harmonization (ICH) Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients; November 2000
- Food and Drug Administration (FDA): 21 CFR Part 11; 21 CFR parts 210 and 211 and 820
- European Medicines Agency (EMA): EudraLex Volume 4, Part 1: Basic Requirements for Medicinal Products; Annex 11: Computerized Systems
- European Medicines Agency (EMA): Guideline on computerized systems and electronic data in clinical trials

4 DEFINITIONS

- 4.1 Acceptance Criteria—The established standard that a system or component must satisfy for a result to be acceptable to the user, customer or other authorized entity.
- 4.2 Change Control Request Form—An official request for prospective approval to implement a change to a validated system.
- 4.3 Commercial Off-the-Shelf (COTS) System — A commercially available electronic system (including hardware or software) that can be purchased from third-party vendors.
- 4.4 Computerized System—The set of hardware, software, peripheral devices, personnel, and documentation needed to perform a specific group of functions to meet a business or regulatory need. A Computer System (defined above) is a component of a Computerized System.



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- 4.5 Computerized Systems Validation (CSV)—A process of establishing and documenting that the specified requirements of a computerized system can be consistently fulfilled from design until decommissioning of the system or transition to a new system. The approach to validation should be based on a risk assessment that takes into consideration the intended use of the system and the potential of the system to affect trial participant protection and the reliability of trial results.
- 4.6 Design Specifications—Document which captures the design of a system or system component to satisfy specified requirements.
- 4.7 Functional Requirements Specification (FRS)— The Functional requirements contain User Requirements and Functional Specifications. It describes the Computerized System in terms of the functions it will perform, and facilities required to meet the User Requirements. They also document the fundamental processes or transformations that the Computerized System performs on inputs to produce outputs.
- 4.8 Process/Business Owner—The person ultimately responsible for the business process or processes being managed. The process owner is responsible for ensuring that the computerized system and its operation is in compliance and fit for intended use in accordance with applicable Standard Operating Procedures (SOPs) throughout its useful life. Responsibility for control of system access should be agreed between process and system owner. In some cases, the process owner also may be the system owner.
- 4.9 Subject Matter Expert (SME)— Individuals with highest level of expertise/competency in skills or performing specialized tasks who can provide specific detailed knowledge or expertise.
- 4.10 Traceability Matrix (TM)— A matrix that maps the specification requirements to the test scripts verifying that the specification was met.
- 4.11 User Requirements Specification (URS)—Document that describes what the equipment or system is supposed to do and is normally written by the user. The UR includes all essential requirements (musts), and if possible, a prioritized list of desirable requirements (should). Requirements should be linked to the PQ, which tests the system in its operating environment, including the associated procedures.
- 4.12 Validation Plan (VP)—A formal document providing a framework that describes the validation process, responsibilities, documentation deliverables, timelines, acceptance criteria and maintenance of the validated state for a given computerized system. The VP encompasses the system's lifecycle (development, implementation, use and retirement).



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4.13 Validation Summary Report (VSR)—A document that summarizes the validation activities, exceptional conditions occurring during testing and the justification for releasing for use or continued use of the system.

5 RESPONSIBILITIES

5.1 Business Owner (BO) or Designee

5.1.1 Defines intended system use and business requirements.

5.1.2 Ensures systems remain appropriate for intended use.

5.2 Quality Assurance (QA) System Validation or Designee

5.2.1 Provides oversight for validation activities where applicable.

5.2.2 Ensures validation documentation and records are maintained appropriately.

5.3 Subject Matter Expert (SME) or Designee

5.3.1 Participate in requirements definition, testing, and operational evaluation activities where applicable.

5.3.2

5.4 Technical Team / Information Technology (IT) CSV or Designee

5.4.1 Supports implementation, maintenance, and testing activities.

5.4.2 Maintains system infrastructure and technical configurations.

6 PROCEDURE

6.1 Computer System Validation Service Request

6.1.1 The Business Owner identifies the potential need for validation and identifies the potential need for validation activities associated with a new system, system change, or new intended use.

6.2 Computerized System Regulatory Assessment

6.2.1 Appropriate personnel perform a regulatory and risk assessment for computerized systems where applicable.



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6.2.2 Systems may be categorized based on complexity, configuration, customization, and regulatory impact to support determination of appropriate validation activities.

6.2.2.1 The 21 CFR Part 11 Electronic Records and Electronic Signatures Assessment—Performed to determine if 21 CFR Part 11 regulations are applicable to the computerized system. The assessment includes determining if the system will create, modify, maintain, archive, retrieve, or transmit electronic records that will be used to satisfy FDA and EU established rules and if the system is appropriate for use of electronic signatures.

6.2.2.2 GxP Assessment—Performed to check if the computer system has a potential to impact on the safety and efficacy of a medicinal product.

6.3 Validation Methodology

6.3.1 Netramind applies a risk-based validation approach informed by GAMP 5 principles where appropriate.

6.3.2 Validation Document Numbering—The documents generated as a part of the computer systems lifecycle will have a unique document number in the format CSV-AAA-XXXX.YY where:

- AAA is the abbreviation of the document (e.g., VP for Validation Plan).
- XXXX is the sequential number of each validation project.
- YY is the version number of the document.
- For example, the document number for Validation Plan for the system is CSV-VP-0001.00.

6.4 Validation Process

6.4.1 Planning Phase

During the planning phase, the following validation activities are performed.



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6.4.1.1 The Business Owner along with appropriate personnel create the Business Requirements Specification (BRS) and the document is reviewed and approved as appropriate.

6.4.1.2 The BRS defines the criteria by which the system will be validated against and includes the following, as applicable:

- a. User Requirements (UR): A User Requirements document will be produced by the business owner. This document will document the system's business and functional requirements from a user's perspective. It includes statements of the features that need to be delivered in the final solution that will satisfy the need, goals and objectives defined in the scope. It may include business process diagrams and flowcharts as applicable. It will also include any applicable regulatory requirements including all GxP user requirements. The requirements are independent of the particular hardware and software that are chosen for a specific system.
 - UR Identifier: Unique requirement number is required for each requirement for an easy reference.
 - Requirements may be prioritized based on business, regulatory, and operational impact.
- b. Functional Specification (FS): A functional Specification (FS) will be produced by the project team if applicable. It will include a description of the system in terms of the functions it will perform to meet the user requirements as defined in the UR. The FS should be detailed enough to ensure that functionality, design constraints, attributes, interfaces, calculations are specified.
 - FS Identifier: Unique requirement number is required for each specification for an easy reference and must map to at least one user requirement.



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- 6.4.1.3 Contract Service Provider/Supplier Assessment and Audit: Vendor or supplier assessments may be performed where appropriate based on system risk and intended use.
- 6.4.1.4 Risk Assessment: IT QA along with SME performs a Risk Assessment for all systems. The Business Owner reviews and approve the assessment. The purpose of the Risk Assessment is to determine the risk associated with implementing the proposed new computer system or update to an existing system.
- 6.4.1.5 Validation Plan: Validation activities and documentation may be planned and documented based on system risk, complexity, and intended use.

Traceability between requirements, testing activities, and implemented functionality may be maintained where appropriate.

- 6.4.1.6 Where vendor/supplier is involved, a review of the vendor/supplier documentation must be performed and documented to ensure that it is consistent with requirements specified within applicable regulatory agencies and Netramind procedures.
- 6.4.1.7 Design Specification: The Design Specification provides detailed information about the actual system and includes the following:
- High-level system overview
 - Development and Testing methodology applied
 - List of hardware and software components
 - Database design, including audit trails
 - Report formats
 - Details of interfaces with other systems and users
 - Security
 - Data Requirements including Data Migration if applicable
- 6.4.1.8 Configuration Specification: The purpose of the Configuration Specification is to provide more detailed specifications. Configuration details relevant to system operation, security, and intended use may be documented where appropriate.



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6.4.2 Validation Testing Phase

Validation and testing activities are performed based on system risk, intended use, and complexity to provide reasonable assurance that systems function as intended.

6.4.2.1 Testing activities may include installation verification, functional testing, user acceptance testing, integration testing, or other verification activities as appropriate.

6.4.2.2 Data Migration Plan: Where applicable, data migration activities are planned and verified to ensure data integrity and completeness are maintained.

6.4.3 System Release

6.4.3.1 Systems may be released for operational use following completion of appropriate validation, testing, approval, and training activities where applicable.

6.4.3.2 Training Materials and Training Users: Written procedures, Training materials or work instructions will be created and approved as applicable. Relevant users are trained on applicable systems and procedures prior to operational use where appropriate.

6.4.3.3 Validation activities and outcomes may be summarized in validation records or reports as appropriate based on system risk and complexity.

6.4.3.4 Update Computer System Inventory: Once the Validation Summary Report is approved, the IT QA prepare and update the computer system inventory that includes:

- Project name
- System name and description in short
- Version number of the system
- Risk Level / Results of assessment
- Business owner



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- System location
- Status of the system

6.5 Operation and Maintenance Phase

6.5.1 Once the system has been released for use the operation and maintenance process begins. During this period, the system is operating normally, and any deviations that occur are documented, examined and corrected. Periodic reviews are performed to ensure that the validated state is maintained.

6.5.1.1 Validation Periodic Review: Periodic reviews may be performed for validated systems based on system risk, complexity, criticality, and rate of change to confirm systems remain appropriate for intended use.

6.6 Change Control

6.6.1 Changes to validated systems are managed through the IT Change Control process. Validation impact assessments are performed where appropriate based on the nature and risk of the change.

6.7 Retirement Phase

6.7.1 Validated systems are to be retired when no longer in Use. This process consists of withdrawal of the system from use and archival of data from the system.

6.7.2 Retirement activities may include decommissioning system components, disposition of data and documentation, and updates to system inventory records where appropriate.

6.7.3 Disposition of the Data and Documentation: Prior to decommissioning of a system, all the data contained within is either destroyed, archived, or migrated to a new system depending on the business need. When a system is decommissioned, all system-related documentation is reviewed and archived based upon contractual agreements and regulatory requirements.

6.7.4 The system Retirement Report is written by the IT CSV and summarizes the results from the execution of the System Retirement Plan.



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7 ATTACHMENTS

- **FRM-012-01 IT Change Control Form**

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8 REVISION HISTORY

Version Number	Date	Summary of Change(s)
01	22-MAY-2026	New Document
[X.0]	[DD-MMM-YYYY]	[Insert a summary of the changes]



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SIGNATURE PAGE

[ADD ADDITIONAL ROWS AS NEEDED]

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